

Green and Sustainable Energy: Emerging Technologies, Global Trends, and Future Directions

A Comprehensive Review (2020-2025)

Hadia Arshad

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Abstract4

The global energy landscape is undergoing an unprecedented transformation driven by the urgent need to decarbonize economies, achieve climate targets set under the Paris Agreement, and fulfill Sustainable Development Goal 7 (SDG 7) on affordable and clean energy. This comprehensive review synthesizes peer-reviewed literature, intergovernmental reports, and policy documents published primarily between 2020 and 2025 to evaluate the current state, emerging innovations, and future directions of green and sustainable energy technologies. The review covers solar photovoltaics (PV) and concentrated solar power (CSP), wind energy (onshore and offshore), green hydrogen production and storage, advanced energy storage systems, smart grid infrastructure, bioenergy, geothermal and ocean energy, and carbon capture, utilization, and storage (CCUS). Key findings indicate that the levelized cost of energy (LCOE) for solar PV and onshore wind has fallen by more than 85% over the past decade, making renewables the cheapest source of new electricity generation in most markets. Perovskite solar cells have achieved certified efficiencies exceeding 26%, approaching the performance of conventional silicon cells. Green hydrogen, produced via water electrolysis powered by renewable electricity, is emerging as a critical energy carrier for hard-to-abate industrial sectors. Concurrently, lithium-ion battery costs have declined to approximately USD 115 per kWh in 2024, enabling large-scale grid storage deployment. Artificial intelligence and machine learning are being increasingly integrated into energy forecasting, demand-response systems, and smart grid management. Despite these advances, significant challenges remain, including grid stability, rare earth material constraints, and persistent energy poverty in developing nations such as Pakistan. This review highlights the necessity of integrated policy frameworks, international cooperation, and technology transfer to accelerate the sustainable energy transition globally.

Keywords: renewable energy, solar photovoltaics, green hydrogen, energy storage, smart grids, wind energy, bioenergy, carbon capture, energy policy, sustainable development

1. Introduction

The global energy system stands at a critical juncture. Approximately 80% of global primary energy consumption is currently derived from fossil fuels—coal, oil, and natural gas—whose combustion releases greenhouse gases (GHGs) responsible for driving anthropogenic climate change. The Intergovernmental Panel on Climate Change (IPCC) has unequivocally stated that limiting global average temperature rise to 1.5°C above pre-industrial levels requires net-zero CO₂ emissions by approximately 2050 and steep reductions in other GHGs.

The Paris Agreement, adopted in 2015 and ratified by 196 parties, established a binding international framework for climate action, with nationally determined contributions (NDCs) guiding each country's decarbonization roadmap. Complementing this, SDG 7 of the United Nations 2030 Agenda calls for ensuring universal access to affordable, reliable, sustainable, and modern energy. As of 2023, however, approximately 685 million people worldwide lacked access to electricity, the majority residing in Sub-Saharan Africa and South Asia. Pakistan, for instance, faces a dual challenge: it must expand energy access to a population exceeding 230 million while simultaneously decarbonizing its energy mix, which remains heavily dependent on imported fossil fuels.

Global renewable energy capacity has expanded rapidly. According to the International Renewable Energy Agency (IRENA), total renewable power capacity reached 3,372 GW at the end of 2022, with solar and wind accounting for the largest shares of new additions. The International Energy Agency (IEA) projects that renewables will supply nearly 90% of global electricity generation by 2050 under a net-zero emissions scenario. Investment in clean energy surpassed USD 1.7 trillion in 2023, with 2024 estimates exceeding USD 2 trillion, surpassing fossil fuel investment for the first time in history.

Despite this progress, the transition is uneven across regions and technologies. High-income nations benefit from mature financial markets, established grid infrastructure, and robust research and development ecosystems. Middle and low-income countries, including Pakistan, often lack the institutional capacity, grid reliability, and capital access needed to deploy renewables at scale. Moreover, the intermittent nature of solar and wind energy necessitates advances in energy storage, grid management, and demand-side flexibility that remain areas of active research.

This review aims to provide a systematic and comprehensive synthesis of developments in green and sustainable energy technologies from 2020 to 2025. It covers the full technology stack from generation to storage to distribution and contextualizes findings within global policy and economic frameworks. Special attention is given to the relevance of these technologies for developing nations, with Pakistan highlighted as a case study of both opportunity and challenge. The structure follows a logical progression from primary energy generation technologies (solar, wind, hydrogen, bioenergy, geothermal, ocean) through enabling infrastructure (storage, smart grids) to cross-cutting themes (CCUS, policy, economics, and future directions).

The methodology employed is a narrative review of peer-reviewed journal articles, intergovernmental reports (IEA, IRENA, IPCC), and policy documents. Databases searched include Web of Science, Scopus, Google Scholar, and the official repositories of IEA and IRENA. Search terms included combinations of "renewable energy," "solar PV," "wind energy," "green hydrogen," "energy storage," "smart grid," "bioenergy," "geothermal," "tidal energy," "carbon capture," and "energy policy," filtered to literature published between 2018 and 2025, with older seminal works included where necessary for context.

2. Solar Energy: Advances and Challenges

2.1 Photovoltaic Efficiency Improvements

Solar photovoltaic technology has experienced remarkable efficiency gains and cost reductions over the past decade, making it the fastest-growing energy technology in history. Crystalline silicon (c-Si) solar cells, which dominate the market with approximately 95% share, have seen commercial module efficiencies rise to 22-24% from laboratory records approaching 26.7% for monocrystalline PERC (Passivated Emitter and Rear Cell) architectures. These improvements arise from advances in surface passivation, light trapping, and contact engineering.

Heterojunction technology (HJT) and tunnel oxide passivated contact (TOPCon) cells represent the current frontier of commercial silicon PV, with HJT modules achieving efficiencies above 23% and demonstrating superior temperature coefficients compared to conventional technologies. Multi-junction concentrator PV cells, using III-V semiconductor materials, have achieved laboratory efficiencies exceeding 47% under concentrated sunlight, though their high cost currently restricts deployment to niche applications such as space power systems.

Table 1: Comparison of Solar Cell Technologies by Efficiency and Maturity (2020–2025)

Technology	Record Lab Efficiency (%)	Commercial Efficiency (%)	Technology Readiness Level	Cost (USD/Wp)
Mono c-Si (PERC)	26.7	22–24	Commercial	0.20–0.30
HJT	26.5	23–24.5	Commercial	0.25–0.35
TOPCon	26.1	22–23.5	Commercial	0.22–0.32
Perovskite (single)	26.1	18–20	Pilot/Emerging	0.10–0.20 (est.)
Perovskite/Si Tandem	33.9	25–28	Pre-commercial	N/A
CdTe (thin film)	22.1	18–19	Commercial	0.20–0.28
CIGS (thin film)	23.6	16–18	Commercial	0.25–0.35

Sources: NREL Best Research-Cell Efficiency Chart (2024–2025); Green et al., *Progress in Photovoltaics* (2024); IRENA Renewable Power Generation Costs (2022)

2.2 Perovskite Solar Cells

Perovskite solar cells (PSCs) have emerged as the most exciting development in PV research over the past decade. Named after the ABX_3 crystal structure (where A = organic/inorganic cation, B = Pb^{2+} or Sn^{2+} , X = halide), perovskites exhibit exceptional optoelectronic properties including high absorption coefficients, long carrier diffusion lengths, and tunable bandgaps. Certified single-junction perovskite efficiencies have risen from below 4% in 2009 to over 26.9% in 2025, an unprecedented rate of improvement.

Perovskite/silicon tandem cells, which combine the high-voltage perovskite top cell with a silicon bottom cell to harvest a broader solar spectrum, have achieved certified efficiencies of 34.85% (LONGi, NREL-certified, 2024), surpassing the Shockley-Queisser single-junction limit of approximately 33.7%. These tandem architectures are widely regarded as the pathway to next-generation commercial solar cells with efficiencies potentially exceeding 30% at moderate cost.

The principal challenges facing PSC commercialization are stability, scalability, and toxicity. Lead-containing perovskites degrade under moisture, heat, and ultraviolet light, limiting operational lifetimes. Encapsulation strategies, compositional engineering (adding Cs^+ , Rb^+ , or mixed halides), and 2D/3D hybrid architectures have improved stability to thousands of hours under accelerated testing. The replacement of lead with less toxic tin or bismuth alternatives remains an active research frontier, though tin-based perovskites currently exhibit lower efficiency and greater instability.

2.3 Concentrated Solar Power

Concentrated Solar Power (CSP) uses mirrors or lenses to focus sunlight onto a receiver, generating high-temperature heat that drives a conventional

thermodynamic cycle. Unlike PV, CSP can store energy thermally for hours, enabling dispatchable solar electricity. Parabolic trough systems, power towers (central receiver), and linear Fresnel reflectors represent the primary CSP configurations.

Global installed CSP capacity reached approximately 7 GW by 2023, concentrated in Spain, the United States, Morocco, and the United Arab Emirates. The Noor Ouarzazate complex in Morocco, one of the world's largest CSP facilities, demonstrates the technology's relevance for North Africa and the Middle East, regions with high direct normal irradiance (DNI) values. Pakistan's Balochistan province similarly possesses high DNI resources that could support CSP development, though no utility-scale CSP projects had been commissioned as of 2025.

Advances in molten salt thermal storage have extended CSP dispatchability to 10–18 hours, enabling round-the-clock solar electricity generation. Research into supercritical CO₂ (sCO₂) power cycles promises higher thermodynamic efficiencies (>50%) at elevated operating temperatures, potentially reducing the LCOE of CSP below USD 0.06/kWh.

2.4 Solar-Grid Integration

The rapid expansion of solar PV capacity poses integration challenges for existing electricity grids. High solar penetration introduces variability, potential voltage instability, and the "duck curve" phenomenon a sharp increase in net load in the evening hours as solar output declines. Grid operators have responded with a suite of technical and market measures, including curtailment management, dynamic line rating, expanded transmission interconnections, and flexible demand response programs.

Power electronics advances, particularly in wide-bandgap semiconductor devices (SiC and GaN), have improved the efficiency and cost of solar inverters while enabling grid-forming inverter capabilities that support voltage and frequency stability at high renewable penetrations. Smart inverters with reactive power control, low-voltage ride-through, and grid-support functions are increasingly mandated by grid codes in Europe, the United States, and China.

3. Wind Energy: Onshore and Offshore Technologies

3.1 Turbine Innovations

Wind energy is the second-largest renewable electricity source globally, with cumulative installed capacity reaching approximately 906 GW by the end of 2022.

Modern utility-scale wind turbines have grown substantially in size and power output, with commercial onshore turbines now rated at 5–7 MW and rotor diameters exceeding 160 m. This scaling trend reduces the LCOE by spreading capital costs over greater energy yields per turbine, lowering infrastructure and maintenance costs per MWh.

Advances in blade aerodynamics, including winglet-tipped blades and flexible smart blades with real-time load control, have improved annual energy yield by 3–8% compared to previous generations. Direct-drive permanent magnet generators eliminate the gearbox historically the most failure-prone component improving reliability and reducing maintenance costs for onshore turbines, while superconducting generator concepts are being explored for even larger offshore machines.

3.2 Floating Offshore Wind

Fixed-bottom offshore wind turbines are limited to water depths below approximately 60 m, restricting deployment areas. Floating offshore wind (FOW) platforms extend viability to depths exceeding 200 m, opening vast deep-water resources in the Pacific Ocean, the Mediterranean, and the Norwegian Sea. Three principal floating platform concepts spar buoys, semi-submersibles, and tension-leg platforms each offer different trade-offs between stability, cost, and installation complexity.

The Hywind Tampen project in Norway, which uses spar-buoy platforms and achieved full commissioning in 2023, demonstrated the technical viability of multi-turbine FOW arrays at commercial scale. By 2025, global FOW capacity remained below 300 MW but with a substantial pipeline exceeding 100 GW under development in Europe, the United States, Japan, South Korea, and Taiwan. Cost reduction pathways for FOW rely on supply chain standardization, serial manufacturing of platform components, and co-deployment with offshore hydrogen production.

3.3 Wind Forecasting with AI/ML

Accurate wind power forecasting is essential for grid operators managing the variability of wind generation. Machine learning (ML) approaches, particularly deep neural networks including Long Short-Term Memory (LSTM) networks and Transformer architectures, have demonstrated superior forecasting accuracy compared to traditional Numerical Weather Prediction (NWP) models at sub-hourly to 48-hour horizons. Ensemble ML models that combine physical and data-driven approaches have achieved day-ahead forecast errors below 5% of installed capacity in well-instrumented systems.

Wake modeling predicting the turbulence and velocity deficits that downstream turbines experience in large wind farms has similarly benefited from ML integration. Physics-informed neural networks (PINNs) that embed fluid dynamics equations into the learning process have shown promise for computationally efficient and physically consistent wake predictions. These advances are directly relevant to Pakistan's wind corridor in Sindh province, where improved forecasting could enhance the integration of the growing onshore wind fleet.

4. Green Hydrogen: Production, Storage, and Applications

4.1 Water Electrolysis Technologies

Green hydrogen produced by splitting water using renewable electricity is increasingly recognized as a versatile energy carrier capable of decarbonizing hard-to-abate sectors including steelmaking, ammonia synthesis, heavy transport, and high-temperature industrial heat. Three primary electrolysis technologies are in various stages of commercial development: Alkaline Electrolysis (AEL), Proton Exchange Membrane electrolysis (PEMEL), and Solid Oxide Electrolysis (SOEL).

Alkaline electrolysis is the most mature and commercially deployed technology, utilizing a liquid potassium hydroxide electrolyte and nickel-based electrodes. It offers lower capital costs and long operational lifetimes but suffers from slower dynamic response, making it less suited to coupling with variable renewables. PEM electrolysis employs a solid polymer membrane, enabling rapid response to variable power inputs, higher current densities, and the production of high-purity hydrogen. Capital costs for PEM electrolyzers have declined significantly, falling below USD 700/kW in 2024, with continued reductions projected as manufacturing scales up. Solid oxide electrolysis, operating at temperatures of 700–1000°C, offers the highest thermodynamic efficiency but remains at lower technology readiness levels.

Table 2: Comparison of Water Electrolysis Technologies for Green Hydrogen Production

Parameter	Alkaline (AEL)	PEM (PEMEL)	Solid Oxide (SOEL)
Operating Temperature (°C)	60–90	50–80	700–1000
Current Density (A/cm²)	0.2–0.4	1.0–3.0	0.3–1.0
System Efficiency (%)	63–71	67–82	74–90 (with heat input)
H ₂ Purity (%)	99.5–99.9	99.99+	99.9+
Capex (USD/kW)	500–1400	700–1400	2000–3000
Dynamic Response	Slow	Fast	Slow-Moderate
Technology Readiness Level	9 (commercial)	8–9 (commercial)	6–7 (pilot)

Sources: Ursua et al., *Proceedings of the IEEE* (2012); Carmo et al., *International Journal of Hydrogen Energy* (2013); IRENA *Green Hydrogen Cost Reduction* (2020); Schmidt et al., *International Journal of Hydrogen Energy* (2017)

4.2 Hydrogen Storage Materials

Efficient and safe hydrogen storage is a key bottleneck in the hydrogen economy. Molecular hydrogen has a very low volumetric energy density at ambient conditions, necessitating compression (to 350–700 bar), liquefaction (at -253°C), or material-based storage. Metal hydrides such as MgH_2 , LaNi_5H_6 , and complex hydrides like NaAlH_4 offer high gravimetric storage capacities but typically require elevated temperatures for hydrogen release. Porous materials, including metal-organic frameworks (MOFs), hydrogen clathrates, and activated carbons, have demonstrated promising adsorption capacities at cryogenic temperatures and are subjects of intense research.

Liquid organic hydrogen carriers (LOHCs), such as dibenzyltoluene (DBT), enable hydrogen transport in existing liquid fuel infrastructure at ambient conditions, with hydrogenation and dehydrogenation conducted at industrial sites. This approach is particularly attractive for long-distance hydrogen export, with Japan's international hydrogen supply chain demonstration involving Australia and Brunei providing real-world validation.

4.3 Fuel Cells

Hydrogen fuel cells convert chemical energy directly to electricity via electrochemical oxidation of hydrogen, with water as the only by-product. Proton exchange membrane fuel cells (PEMFCs) dominate transportation applications due to their high power density, rapid start-up, and operation at near-ambient temperatures. Solid oxide fuel cells (SOFCs), operating at $600\text{--}1000^{\circ}\text{C}$, are better suited to stationary power generation, particularly in combined heat and power (CHP) configurations where waste heat is utilized. The global fuel cell market was valued at approximately USD 4.8 billion in 2023 and is projected to grow at a compound annual growth rate (CAGR) of $\sim 20\%$ through 2030, driven by the transportation and industrial sectors.

4.4 Green Ammonia as Hydrogen Carrier

Ammonia (NH_3) has emerged as a compelling hydrogen carrier due to its high hydrogen content (17.6 wt%), established global production and distribution infrastructure, and potential use as a zero-carbon fuel in its own right. Green ammonia synthesis combining green hydrogen with nitrogen via the Haber-Bosch process powered by renewable electricity could reduce the approximately 1.8% of global CO_2 emissions attributable to conventional ammonia production. Several

large-scale green ammonia projects exceeding 1 GW of electrolyzer capacity were under development in Saudi Arabia, Australia, Oman, and Chile as of 2024.

Pakistan produces approximately 1.8 million tonnes of ammonia annually for fertilizer production, representing a significant potential market for green ammonia as natural gas is phased down. The country's renewable energy potential estimated at over 170,000 MW of solar and wind combined positions it as a prospective green ammonia exporter, though significant policy and infrastructure development would be required.

5. Energy Storage Systems

5.1 Lithium-Ion Batteries

Lithium-ion batteries (LIBs) have become the dominant technology for both transportation electrification and stationary grid storage, driven by dramatic cost reductions pack costs fell from approximately USD 1,100/kWh in 2010 to approximately USD 115/kWh in 2024 (BloombergNEF 2024 Battery Price Survey). LIBs operate via the intercalation of lithium ions between a graphite anode and a transition metal oxide cathode (NMC, NCA, or LFP) through a liquid electrolyte.

Lithium iron phosphate (LFP) chemistry has gained substantial market share for stationary storage and entry-level electric vehicles due to its superior thermal stability, long cycle life (>4,000 cycles), and lower cost, despite a lower energy density compared to NMC. Global LIB manufacturing capacity surpassed 1 TWh/year in 2023 and is projected to reach over 6 TWh/year by 2030, primarily driven by gigafactory investments in China, the United States, and Europe.

5.2 Solid-State Batteries

Solid-state batteries (SSBs) replace the liquid electrolyte with a solid ionic conductor, promising higher energy densities, improved safety (elimination of flammable electrolyte), and potentially longer cycle lives. Key solid electrolyte families include sulfide-based (e.g., $\text{Li}_6\text{PS}_5\text{Cl}$), oxide-based (e.g., LLZO, NASICON-type), and polymer electrolytes, each with distinct trade-offs in ionic conductivity, processability, and stability. Major automakers including Toyota, QuantumScape (Volkswagen), and Samsung SDI have announced target timelines of 2027–2030 for commercial SSB electric vehicle deployment. Significant challenges remain in achieving sufficient ionic conductivity at room temperature, managing interfacial resistance between electrode and electrolyte, and developing scalable manufacturing processes.

5.3 Flow Batteries

Vanadium redox flow batteries (VRFBs) store energy in dissolved vanadium electrolyte solutions circulated through an electrochemical cell. Key advantages include decoupled power and energy capacity (enabling independent optimization), very long cycle life (>20,000 cycles), and no cross-contamination between electrolytes. VRFBs are well-suited to long-duration energy storage (LDES) applications of 4–12 hours, where their energy cost advantage over LIBs becomes apparent at large scale. Installed VRFB capacity exceeded 1 GW globally by 2024, with China accounting for the majority of deployments. Alternative flow chemistries including zinc-bromine, iron-air, and organic (anthraquinone-based) systems are under active development as lower-cost alternatives to vanadium.

5.4 Supercapacitors

Supercapacitors (also termed ultracapacitors or electrochemical double-layer capacitors, EDLCs) store energy electrostatically at the electrode-electrolyte interface, enabling extremely high power densities and cycle lives exceeding one million cycles. While their energy density (5–15 Wh/kg) is far below that of batteries, supercapacitors excel in applications requiring rapid charge-discharge cycles, such as regenerative braking in transportation, power quality conditioning, and grid frequency regulation. Hybrid supercapacitors, which combine faradaic (battery-like) and capacitive storage mechanisms, are being developed to bridge the energy density gap.

5.5 Thermal Energy Storage

Thermal energy storage (TES) encompasses a diverse set of technologies including sensible heat (molten salts, water tanks), latent heat (phase change materials, PCMs), and thermochemical storage. TES is particularly relevant to CSP plants and district heating/cooling systems, where low-cost thermal storage enables temporal shifting of energy use. Molten salt two-tank systems in commercial CSP plants provide storage at costs as low as USD 15–25/kWh (thermal), substantially lower than electrochemical alternatives.

Table 3: Comparison of Grid-Scale Energy Storage Technologies

Technology	Energy Density (Wh/kg)	Power Density (W/kg)	Cycle Life	LCOS (USD/MWh)	Discharge Duration
Lithium-ion (LFP)	120–180	500–2000	>4,000	100–200	1–4 h
Vanadium Redox Flow	15–25	50–150	>20,000	150–300	4–12 h
Solid-State (emerging)	300–500 (est.)	>2000	>5,000 (proj.)	TBD	1–4 h
Lead-Acid	30–50	75–300	500–1500	200–400	2–8 h

Supercapacitor	5–15	1000–10000	>1,000,000	High (power apps)	Seconds–minutes
Pumped Hydro	~0.3 Wh/L (system)	Low	50+ years	100–200	6–24 h

Sources: BloombergNEF Electric Vehicle Outlook 2024; Nykvist& Nilsson, *Nature Climate Change* (2015); Skyllas-Kazacos et al., *Journal of the Electrochemical Society* (2011); Mongird et al., *PNNL* (2020); Lovegrove et al., *Solar Energy Materials and Solar Cells* (2021)

6. Smart Grids and AI in Energy Management

6.1 IoT-Enabled Smart Grids

The traditional electricity grid, designed for centralized, unidirectional power flow from large generators to passive consumers, is being transformed into a smart grid: a bidirectional, digitally instrumented network capable of accommodating distributed energy resources (DERs), electric vehicles, and responsive demand. The Internet of Things (IoT) provides the communication backbone of smart grids, connecting sensors, smart meters, automated switches, and distributed intelligence across generation, transmission, and distribution infrastructure. Advanced metering infrastructure (AMI), which enables two-way communication between utilities and customers, had been deployed to approximately 1.1 billion endpoints globally by 2024, enabling real-time consumption data collection and remote management.

In Pakistan, the National Electric Power Regulatory Authority (NEPRA) has initiated smart metering pilots in major urban centers including Karachi and Lahore as part of the country's Power Sector Reform Roadmap, though deployment remains limited relative to the scale of the national grid. The potential for smart grid infrastructure to reduce transmission and distribution losses which average approximately 18% in Pakistan, far above the global average of ~8% represents a significant economic opportunity.

6.2 Demand-Response Systems

Demand response (DR) programs incentivize electricity consumers to reduce, shift, or reshape their load in response to price signals or reliability needs, providing system flexibility without additional generation capacity. Economic DR programs use time-of-use pricing, real-time pricing, or peak demand charges to encourage voluntary load shifting, while emergency DR programs dispatch committed load reductions to prevent grid instability. Automation is increasingly central to DR: smart building management systems, vehicle-to-grid (V2G) systems, and smart appliances can execute DR responses without requiring manual consumer action.

6.3 Machine Learning for Energy Prediction

Machine learning has become indispensable in modern energy management, applied to load forecasting, renewable generation forecasting, energy price prediction, fault detection, and asset optimization. Gradient boosting algorithms (XGBoost, LightGBM) and deep learning architectures (LSTM, Transformer, Temporal Convolutional Networks) consistently outperform traditional statistical methods such as ARIMA for energy forecasting tasks. A particularly active research area is the use of federated learning, which allows forecasting models to be trained across distributed smart meters without centralizing sensitive consumer data, preserving privacy while leveraging large datasets.

6.4 Blockchain in Energy Trading

Blockchain technology offers a decentralized, transparent, and tamper-resistant ledger that may underpin next-generation peer-to-peer (P2P) energy trading platforms, enabling prosumers (consumers who also produce energy) to sell excess renewable generation directly to neighbors without intermediary utilities. Smart contracts self-executing code deployed on blockchain platforms can automate settlement, enforce trading rules, and ensure compliance with grid codes. Pilot P2P trading projects in Australia (PowerLedger), Brooklyn (TransActive Grid), and Germany (Share&Charge) have demonstrated technical feasibility, though regulatory frameworks in most jurisdictions do not yet fully accommodate decentralized energy trading models.

7. Bioenergy and Biomass Conversion

7.1 Lignocellulosic Biofuels

Bioenergy energy derived from biomass provides approximately 10% of global primary energy and remains the largest single renewable energy source globally, particularly for heating and cooking in rural developing regions. Second-generation (advanced) biofuels derived from non-food lignocellulosic biomass including agricultural residues, energy grasses, and forest biomass offer significantly improved sustainability credentials compared to first-generation food-crop-based biofuels. The conversion of lignocellulosic biomass requires pretreatment to disrupt the recalcitrant cell wall structure, enzymatic hydrolysis to release fermentable sugars, and microbial fermentation to produce ethanol or other products.

Thermochemical conversion routes including gasification to syngas and subsequent Fischer-Tropsch synthesis enable the production of "drop-in" hydrocarbon fuels compatible with existing aviation and shipping infrastructure. Sustainable aviation fuel (SAF) from biomass is being actively pursued under

ICAO's CORSIA framework, with the IEA projecting SAF production of 450 million litres by 2025, rising substantially thereafter.

Pakistan generates approximately 50 million tonnes of agricultural residues annually from wheat straw, rice husk, and sugarcane bagasse, representing a substantial feedstock base for bioenergy and biofuel production. Utilization of these residues for bioenergy, rather than open-field burning a common practice that causes severe air pollution could simultaneously improve energy supply and air quality.

7.2 Anaerobic Digestion and Biogas

Anaerobic digestion (AD) converts organic waste agricultural residues, municipal solid waste (MSW), wastewater sludge, and animal manure into biogas (typically 50–70% CH₄) through microbial decomposition in the absence of oxygen. Biogas can be used directly for heat and power generation or upgraded to biomethane (>97% CH₄) for injection into natural gas networks or use as a vehicle fuel. Digestate the nutrient-rich residue from AD serves as an organic fertilizer, enhancing nutrient cycling and reducing dependence on synthetic fertilizers.

Global installed biogas capacity exceeded 19 GW electrical equivalent by 2023, with Europe particularly Germany, Italy, and the UK accounting for the largest share. The potential for AD-based biogas in Pakistan and South Asia is significant given the large livestock population and agricultural waste streams, though development is constrained by fragmented waste collection systems and limited technical capacity.

7.3 Algae-Based Bioenergy

Microalgae represent a third-generation biofuel feedstock with several potential advantages over terrestrial biomass: very high photosynthetic efficiency, rapid growth rates, cultivation on non-arable land using wastewater or saline water, and the capacity to accumulate substantial lipid content suitable for biodiesel production. Some species accumulate lipids exceeding 50% of dry cell weight under nitrogen limitation. However, significant technical and economic barriers remain, including high capital and operating costs for photobioreactors or open raceway ponds, energy-intensive dewatering and lipid extraction, and the challenge of achieving high lipid productivity simultaneously with high biomass productivity.

Life cycle analyses suggest that microalgae biofuels are currently not competitive with petroleum on a cost basis, though they may achieve competitiveness with sustained research investment and the valorization of co-products (proteins, pigments, exopolysaccharides) under a biorefinery model.

8. Geothermal and Ocean Energy

8.1 Enhanced Geothermal Systems

Conventional hydrothermal geothermal energy exploits naturally occurring steam or hot water reservoirs near tectonic boundaries, limiting geographical applicability. Enhanced Geothermal Systems (EGS), also termed engineered or deep geothermal systems, expand the resource base by creating artificial permeability in hot dry rock formations through hydraulic stimulation. EGS has the theoretical potential to provide >100 GWe of baseload renewable electricity in the United States alone, according to the Massachusetts Institute of Technology's landmark "The Future of Geothermal Energy" assessment. The Fervo Energy project in Utah, which began commercial power delivery in 2024 using horizontal drilling techniques adapted from the oil and gas industry, is widely regarded as a significant milestone for EGS commercialization.

Key challenges for EGS include induced seismicity associated with hydraulic stimulation (as experienced in Basel, Switzerland in 2006 and Pohang, South Korea in 2017), parasitic energy consumption for fluid pumping, and uncertainties in subsurface permeability characterization. Next-generation closed-loop geothermal systems, which circulate working fluid through sealed wellbores without open-loop fracturing, may mitigate induced seismicity risk while still accessing deep heat resources.

8.2 Tidal and Wave Energy Converters

Ocean energy encompasses tidal stream, tidal barrage, wave energy, salinity gradient (osmotic), and ocean thermal energy conversion (OTEC) technologies. Tidal stream turbines, resembling underwater wind turbines, extract kinetic energy from tidal currents in straits and channels with high current velocities. The MeyGen project in the Pentland Firth, Scotland, represents the world's first utility-scale tidal stream array. Wave energy converters (WECs) exploit the oscillatory motion of ocean surface waves, with point absorbers, oscillating wave surge converters, and oscillating water columns among the main device types.

Global ocean energy capacity remains modest at approximately 500 MW installed (mostly tidal barrages such as the La Rance plant in France), reflecting the technological immaturity of tidal stream and wave devices, their exposure to harsh marine environments, and high capital costs. IRENA projects ocean energy potential of over 300 GW by 2050 under favorable policy and technological scenarios, concentrated in coastal regions of Europe, Asia, and the Americas.

9. Carbon Capture, Utilization, and Storage (CCUS)

9.1 Direct Air Capture

Carbon Capture, Utilization, and Storage (CCUS) encompasses technologies that capture CO₂ at the point of emission (from power plants or industrial facilities) or directly from the atmosphere (Direct Air Capture, DAC), followed by either permanent geological storage or conversion to useful products. DAC uses chemical sorbents or liquid solvents to selectively bind atmospheric CO₂ (at ~420 ppm), with heat or electricity then used to release the captured CO₂ for compression and storage or utilization.

The Orca plant (12,000 tonnes CO₂/year) operated by Climeworks in Iceland and the Stratos plant (100,000 tonnes CO₂/year, the world's largest) commissioned by Occidental Petroleum in Texas in 2024 demonstrate the technical feasibility of DAC at increasing scale. Current DAC costs of USD 300–1,000 per tonne CO₂ remain far above the USD 100–150/tonne range estimated necessary for broad commercial viability, but learning rates suggest significant cost reduction potential as deployment scales.

9.2 CO₂ Utilization in Fuels and Chemicals

CO₂ utilization converting captured carbon dioxide into value-added products can improve the economics of CCUS by generating revenue streams that offset capture costs. Key utilization pathways include: conversion to methanol or synthetic natural gas via catalytic hydrogenation with green hydrogen (Power-to-X); mineralization into building materials (supplementary cementitious materials); enhanced oil recovery (CO₂-EOR); and production of polycarbonate polymers. Power-to-X pathways couple CCUS with renewable electricity and green hydrogen, creating circular carbon-neutral fuel cycles particularly relevant for aviation, shipping, and chemical feedstocks.

9.3 Integration with Renewable Energy

The integration of CCUS with renewable energy, particularly in bioenergy with carbon capture and storage (BECCS) configurations, can generate negative emissions removing more CO₂ from the atmosphere than the overall process emits. BECCS features prominently in IPCC climate mitigation scenarios that limit warming to 1.5°C, though its large-scale deployment raises concerns about land use competition, water consumption, and biodiversity impacts. The IEA's net-zero scenario projects 980 Mt of CO₂ captured by CCUS per year by 2050, requiring a roughly 35-fold scale-up from current capacity of approximately 45 Mt/year.

10. Sustainable Energy Policy and Economics

10.1 Levelized Cost of Energy Trends

The Levelized Cost of Energy (LCOE) the lifetime cost of an energy project divided by its total energy output is the standard metric for comparing the economic competitiveness of different generation technologies. IRENA's annual renewable cost reports document dramatic LCOE reductions across all major renewable technologies over the past decade. Global weighted-average LCOEs for utility-scale solar PV fell from USD 0.381/kWh in 2010 to USD 0.049/kWh in 2022 an 87% reduction; onshore wind fell from USD 0.102/kWh to USD 0.033/kWh a 68% reduction. These reductions reflect economies of scale, manufacturing learning curves, technology improvements, and increased competition in project development and financing.

By 2022, solar PV and onshore wind were the cheapest sources of new electricity generation in most regions of the world, with their LCOEs substantially below those of new coal and gas plants even without carbon pricing. Offshore wind LCOE, while significantly higher than onshore at approximately USD 0.081/kWh in 2022, has also declined sharply and is projected to converge with onshore wind as floating offshore technology matures.

Table 4: Global Weighted-Average LCOE by Renewable Technology (2010 vs. 2022, USD/kWh)

Technology	LCOE 2010 (USD/kWh)	LCOE 2022 (USD/kWh)	% Reduction
Solar PV (utility-scale)	0.381	0.049	-87%
Concentrating Solar Power	0.346	0.118	-66%
Onshore Wind	0.102	0.033	-68%
Offshore Wind	0.188	0.081	-57%
Bioenergy	0.076	0.066	-13%
Geothermal	0.049	0.057	+16%*

**Slight increase reflects rising development costs for remaining conventional hydrothermal resources.*

Source: IRENA Renewable Power Generation Costs in 2022

10.2 Renewable Energy Investment Flows

Global investment in the clean energy transition encompassing renewable power, grid infrastructure, energy efficiency, clean transportation, and carbon capture reached approximately USD 1.74 trillion in 2023, with 2024 estimates exceeding USD 2 trillion, surpassing fossil fuel investment for the first time. Solar power attracted the largest share at approximately USD 380 billion, driven by utility-scale and distributed installations across China, Europe, and the United States. China alone accounted for approximately USD 758 billion in clean energy investment in 2023, over 40% of the global total.

Investment in developing nations outside China, however, remains inadequate. Sub-Saharan Africa and South and Southeast Asia collectively attracted less than 15% of global clean energy investment despite hosting over 60% of the world's energy-poor population. Pakistan received approximately USD 2.3 billion in clean energy investment in 2023, primarily in solar and wind power, but faces challenges in attracting international capital due to currency risk, political uncertainty, and circular debt in the power sector.

10.3 Policy Frameworks

Policy frameworks are the principal driver of renewable energy deployment, and their design has evolved significantly in the 2020–2025 period. The European Union's Fit for 55 package and REPowerEU initiative accelerated renewable targets to 45% by 2030 (up from the original 32%), with binding national plans for member states. The United States Inflation Reduction Act (IRA), signed into law in August 2022, allocated approximately USD 369 billion in clean energy and climate provisions over ten years, primarily through production tax credits (PTCs) and investment tax credits (ITCs) for solar, wind, batteries, hydrogen, and CCUS. The IRA is widely regarded as the most significant climate legislation in US history and has triggered a wave of domestic clean energy manufacturing investment.

Pakistan's Renewable Energy Policy 2020 established targets of 30% renewable electricity by 2030 and 60% by 2030 in updated NDCs submitted in 2021. The Renewable Energy Auction Policy (REAP) introduced competitive procurement for renewable energy projects, replacing the previous upfront tariff regime. Pakistan's Alternative Energy Development Board (AEDB) has been central to facilitating renewable energy project development, though challenges including currency depreciation, the energy circular debt crisis, and grid capacity limitations have constrained progress. The country's solar potential is among the highest globally, with annual solar irradiation exceeding 5.5 kWh/m²/day in southern and western regions, and a rapid uptake of distributed rooftop solar in urban areas has occurred even in the absence of robust net-metering frameworks.

11. Challenges and Future Perspectives

11.1 Grid Stability and Flexibility

As the share of variable renewable energy (VRE) primarily solar PV and wind increases in national electricity systems, grid stability becomes an increasingly complex challenge. High VRE penetrations require substantial flexibility from the rest of the system, provided by a combination of energy storage, flexible thermal generation, demand response, grid interconnection, and sector coupling. The displacement of synchronous generators (coal, gas, nuclear) by inverter-based

resources removes inertia from the grid, potentially destabilizing frequency following disturbances a concern that is actively being addressed through grid-forming inverter technologies and synthetic inertia services.

Curtailment of renewable energy the deliberate reduction of output below available generation capacity due to grid constraints remains a significant source of waste and economic inefficiency. China curtailed approximately 17 TWh of solar and wind in 2023 despite significant storage deployment, while curtailment in parts of Texas, California, and the UK also exceeded 5% of available renewable generation. Long-duration energy storage and expanded transmission infrastructure are widely cited as the primary remedies.

11.2 Land Use and Resource Constraints

Large-scale deployment of solar and wind requires significant land areas, raising concerns about conflicts with agriculture, biodiversity, and local communities. Agrivoltaics the co-location of solar panels with agricultural activities is emerging as a promising approach to reduce land-use trade-offs, with demonstrated benefits including reduced crop water consumption, protection from extreme weather events, and income diversification for farmers. Floating solar (floatovoltaics) on reservoirs, irrigation canals, and aquaculture ponds similarly avoids direct land-use competition while potentially reducing evaporation.

The clean energy transition is also highly material-intensive. Solar PV requires silver (for electrical contacts), indium (for CIGS), and tellurium (for CdTe); wind turbines require large quantities of rare earth elements (neodymium, dysprosium) for permanent magnet generators; and batteries depend critically on lithium, cobalt, manganese, nickel, and copper. The IEA has warned that projected demand for critical minerals could exceed current production capacity for lithium (by a factor of 42×), cobalt (21×), and nickel (19×) under a net-zero emissions scenario, highlighting the need for circular economy strategies, substitution research, and diversified supply chains.

11.3 Energy Poverty in Developing Nations

Despite global progress, energy poverty persists as a profound development challenge. Approximately 685 million people lack electricity access, and 2.1 billion rely on traditional biomass for cooking, with severe health consequences from indoor air pollution. Distributed renewable energy solutions particularly solar home systems (SHS), solar lanterns, and mini-grids have demonstrated effectiveness in extending energy access to off-grid communities across Sub-Saharan Africa and South Asia. Mini-grids powered by solar PV and battery storage, potentially hybridized with small-scale biogas, are increasingly cost-competitive with diesel generators for rural electrification.

Pakistan's energy poverty challenge is distinct: the country has near-universal grid connectivity but chronic reliability problems, with average load-shedding of 6–12 hours per day in rural areas. The economic cost of power outages is estimated at 2–3% of GDP annually. Affordable, distributed solar PV systems, increasingly deployed as "solar microgrids" by cooperative arrangements, are providing a pragmatic near-term solution in areas underserved by the national grid.

11.4 Research Gaps

Several critical research gaps merit emphasis. The fundamental science of perovskite degradation mechanisms under real-world conditions remains incompletely understood, hindering the rational design of stable encapsulation and passivation strategies. Hydrogen storage at high gravimetric and volumetric density under ambient or near-ambient conditions a "holy grail" of materials chemistry lacks a practical solution. Long-duration energy storage (LDES) technologies capable of storing electricity for days to weeks at low cost (below USD 20/kWh) do not yet exist at commercial scale. The social science dimensions of the energy transition equitable distribution of costs and benefits, just transition frameworks for fossil fuel workers, and community acceptance of renewable energy projects require interdisciplinary research that has lagged behind technical innovation.

12. Conclusion

This comprehensive review has surveyed the rapidly evolving landscape of green and sustainable energy technologies across the full spectrum from generation to storage to distribution and utilization, with particular attention to developments between 2020 and 2025. The overarching conclusion is that the clean energy transition is technically feasible, increasingly economically competitive, and politically mainstreamed across most major economies, but that the pace and breadth of deployment must accelerate dramatically to meet the 1.5°C climate target.

Solar photovoltaics and wind energy have established themselves as the cheapest sources of new electricity generation globally, with continued cost reductions expected. Perovskite solar cells and their tandem configurations with silicon represent the most promising near-term pathway to efficiencies above 30% at low cost, contingent on resolving stability and scale-up challenges. Green hydrogen is emerging as an essential energy carrier for decarbonizing hard-to-abate sectors, with PEM electrolysis costs declining rapidly and major deployment underway in multiple regions. Advanced energy storage technologies from next-generation lithium-ion to solid-state batteries, flow batteries, and long-duration thermal

storage are critical enablers of high-VRE electricity systems. Smart grids, underpinned by IoT sensors, machine learning algorithms, and increasingly blockchain-enabled transactive platforms, are transforming the management of complex, distributed energy systems. Bioenergy, particularly lignocellulosic biofuels and AD-based biogas, plays an important complementary role, especially in developing nations with abundant biomass resources.

For nations like Pakistan, the energy transition simultaneously addresses energy security, economic development, climate commitments, and air quality improvement. The country's immense solar and wind resources, large agricultural biomass potential, and nascent green hydrogen ambitions position it well to benefit from the global clean energy transition provided that policy stability, grid investment, institutional capacity, and international technology transfer are secured. Regional cooperation including with neighboring Afghanistan, India, Iran, and Central Asian states could unlock transformative cross-border renewable energy exchange, as envisioned under the CASA-1000 transmission project and regional energy integration frameworks.

The research community, policymakers, financiers, and civil society must work in concert to translate the remarkable technological progress of recent years into universal energy access, economic prosperity, and a stable climate. The tools are increasingly available; the imperative is urgent; and the decade to 2035 will be decisive.

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